Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/06/2024
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
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City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2025 per NSSC information (Ed., 12/3/24) NOTE: End date changed to 04/30/2024 per NSSC information (Ed., 11/6/23) NOTE: End date changed to 04/30/2023 per D. Stockton/GRC (Ed., 6/6/22) NOTE: End date changed to 06/11/2022 per NSSC information (Ed., 5/17/22) NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 (normal flames) and Round 2 (inverse flames) of Flame Design testing and are processing and analyzing the data. We are running Sphdif simulations and publishing the findings. Three journal papers were published this year and one other was submitted. One conference paper was published and presented. Kendyl Waddell defended and published his Ph.D. dissertation. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames. February 2024 Update per PI. See Bibliography for additional information. (Ed., 2/15/24) Despite a range of conditions, all flames extinguish at a critical temperature of 1130 K, a fuel-based mass flux of 0.2 g/m2-s for normal flames, and an oxygen-based mass flux of 0.68 g/m2-s for inverse flames. With this information, we have developed a simple equation to estimate the flame size (i.e., location of peak temperature) at extinction for any atmospheric-pressure ethylene spherical diffusion flame given only the reactant mass flow rate. Flame growth, which ultimately leads to radiative extinction if the critical extinction point is reached, is attributed to the natural development of the diffusion limited system as it approaches steady state and the reduction in the transport properties as the flame temperature drops due to increasing flame radiation with time (radiation-induced growth). Reference: Irace PH, Waddell KA, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst. 2023 Jan 1;39(2):1905-13. https:// Despite a narrow range of attainable conditions, flames that exhibit steady-state behavior are observed aboard the International Space Station (ISS) for up to 870 seconds, even with the constraint of a finite boundary. Steady-state flames are simulated using the numerical model for over 100,000 seconds. Reference: Irace PH, Waddell KA, Constales D, Kim M, Yablonsky G, Sunderland PB, Axelbaum RL. "On the existence of steady-state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst. 2023 Jan 1;39(2):1721-9. https://
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Frolov SM, Ivanov VS, Frolov FS, Vlasov PA, Axelbaum R, Irace PH, Yablonsky G, Waddell K. "Soot formation in spherical diffusion flames." Mathematics. 2023 Jan 4;11(2):261. https://doi.org/10.3390/math11020261 , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Waddell KA, Constales D, Sunderland PB, Axelbaum RL. "Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar." Proc. Combust. Inst. 2023 Jan 1;39(2):1905-13. https://doi.org/10.1016/j.proci.2022.07.043 [NOTE: previously reported in May 2022 as "in revision."] , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Waddell KA, Constales D, Kim M, Yablonsky G, Sunderland PB, Axelbaum RL. "On the existence of steady-state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss." Proc. Combust. Inst. 2023 Jan 1;39(2):1721-9. https://doi.org/10.1016/j.proci.2022.07.049 [NOTE: previously reported in May 2022 as "in revision."] , Jan-2023
Articles in Peer-reviewed Journals	Irace PH, Gopan A, Axelbaum RL. "An investigation of thermal radiation from laminar diffusion flames in a tri-coflow burner with central oxygen." Combustion and Flame. 2022 Aug 1;242:112158. https://doi.org/10.1016/j.combustflame.2022.112158 , Aug-2022
Articles in Peer-reviewed Journals	Krishnan S, Kumfer BM, Wu W, Li J, Nehorai A, Axelbaum RL. "An approach to thermocouple measurements that reduces uncertainties in high-temperature environments." Energy & Fuels. 2015 Mar 30;29(5):3446-55. https://doi.org/10.1021/acs.energyfuels.5b00071 , Mar-2015

Dissertations and Theses	Waddell K. "Near-limit spherical diffusion flames and cool diffusion flames." Dissertation, University of Maryland, May 2023. , May-2023
Papers from Meeting Proceedings	Waddell KA, Yablonsky G, Constales D, Axelbaum RL, Sunderland PB. "The kinetics and warm flame chemistry associated with radiative extinction of spherical diffusion flames." 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. Abstracts. 13th U.S. National Combustion Meeting, College Station, Texas, March 19-22, 2023. , Mar-2023